

Modeling with Sine and Cosine (5.3)

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: $y = 10$

_____ A sinusoid has maximum 10 and minimum 2. Find its amplitude.

ANSWER: 12

_____ A sinusoid has maximum 20 and minimum -4 . Find its midline.

ANSWER: $y = 6$

_____ Find b so that $y = \sin(bx)$ has period 4.

ANSWER: 5

_____ Find b so that $y = \cos(bx)$ has period 2π .

ANSWER: 4 # _____ A sinusoid has maximum 10 and minimum 2. Find its midline.	ANSWER: 6 # _____ A sinusoid has maximum 15 and minimum 5. Find its amplitude.
ANSWER: $\frac{\pi}{6}$ # _____ A sinusoid has maximum 20 and minimum -4. Find its amplitude.	ANSWER: 1 # _____ A sinusoid has maximum 15 and minimum 5. Find its midline.
ANSWER: $\frac{\pi}{2}$ # _____ Find b so that $y = \sin(bx)$ has period 12.	ANSWER: $y = 8$ # _____ Find the period of a sinusoid with $b = \frac{\pi}{3}$.